FUNDAMENTAL THEOREM OF CALCULUS QUIZ

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Name: Solutions

1. Evaluate the indefinite integral $\int x^2 - x + \cos(x) - \sec^2(x)\,\mathrm{d}x$

$$\frac{1}{3}x^3 - \frac{1}{2}x^2 + \sin(x) - \tan(x) + C$$

2. Given $f'(x) = 3x^2 + 2x + 1$ and $f(2) = 15$, find $f(x)$.

$$f(x) = \int f'(x)\,dx = 3\left(\tfrac{1}{3}x^3\right) + 2\left(\tfrac{1}{2}x^2\right) + x + C = x^3 + x^2 + x + C$$

$$15 = f(2) = 8 + 4 + 2 + C = 14 + C$$

$$\Rightarrow C = 15 - 14 = 1$$

$$\Rightarrow f(x) = x^3 + x^2 + x + 1.$$